

西方经济学

Part 4 Business Cycle Theory

THE ECONOMY IN THE SHORT RUN

Lecture 4C The Supply Side

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(1) M12, M13.3; S5, 6.3. ¹

(2) 其他资料: [GM]

- INTRODUCTION TO ECONOMIC FLUCTUATIONS
- AGGREGATE SUPPLY AND THE SHORT-RUN TRADEOFF BETWEEN INFLATION AND UNEMPLOYMENT



¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

学习目标

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- (1) 掌握 AS 曲线的理论。
- (2) 掌握凯恩斯主义分析框架：AS-AD。
- (3) 理解理性预期在宏观经济政策中的作用。
- (4) 领会马工程教材精神。



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Outline

1 Theories of Aggregate Supply

- The Shape of Aggregate Supply Curve
- The Sticky-Wage Model
- The Standard AS-AD Model

2 Inflation and Unemployment

- Two Causes of Inflation
- Okun's Law
- The Phillips Curve
- Rational Expectations
- Disinflation with Pain
- Disinflation without Pain
- Taylor Rule

3 Appendix: Mankiw

- The Workers' Price-Misperceptions Model
- The Sticky-Price Model
- The Imperfect-Information Model

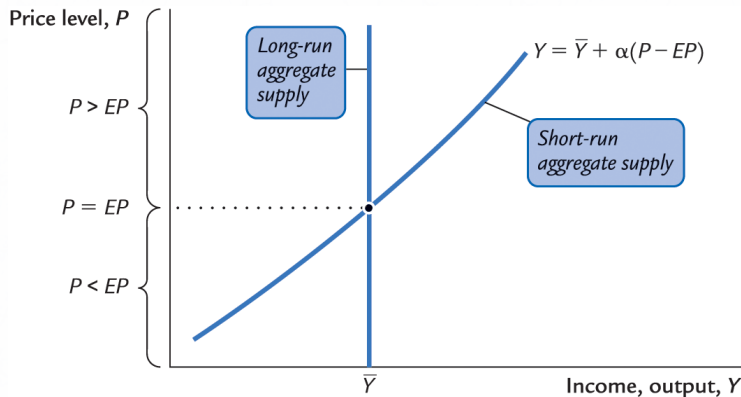
4 马工程教材疑难重点

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The Shape of Aggregate Supply Curve



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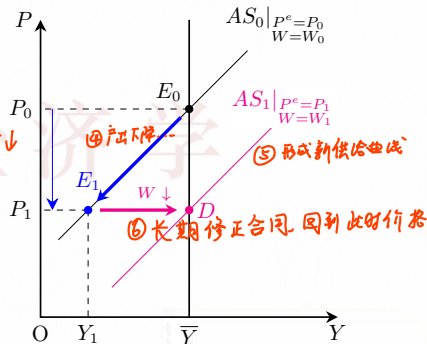
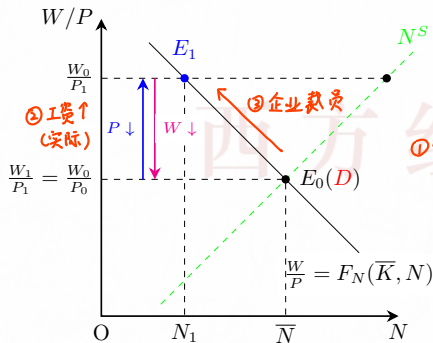
$$Y_t = \bar{Y} + \alpha \cdot (P_t - \mathbb{E}_{t-1}[P_t]), \quad \alpha \in [0, +\infty]$$

$$\Rightarrow P_t = \mathbb{E}_{t-1}[P_t] + \frac{1}{\alpha}(Y_t - \bar{Y})$$



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The Sticky-Wage Model (Gray, 1976; Fischer, 1977)



Setting the nominal wage:

The real wage:

In the short run:

In the long run:

$$W = w^* \cdot P^e,$$

$$\frac{W}{P} = w^* \cdot \frac{P^e}{P}.$$

$$P_0 \searrow P_1 \Rightarrow \bar{N} \searrow N_1, \bar{Y} \searrow Y_1.$$

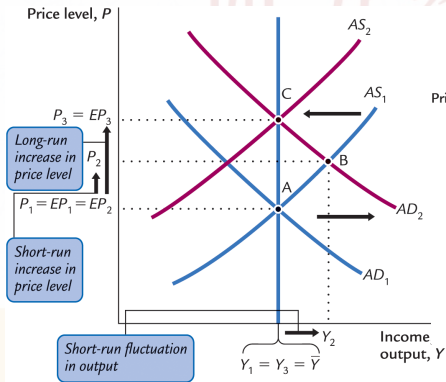
$$W_0 \searrow W_1 \Rightarrow N_1 \nearrow \bar{N}, Y_1 \nearrow \bar{Y}.$$

$$\bar{Y} = F(\bar{K}, \bar{N}), \quad Y_1 = F(\bar{K}, N_1).$$

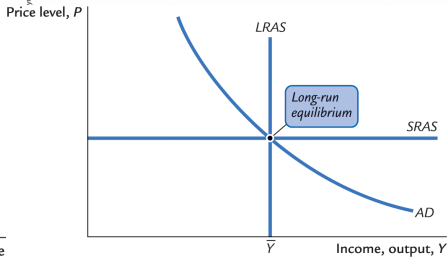
The Standard AS-AD Model

The Short-Run and Long-Run Effects of Monetary or Fiscal Expansion

The Standard AS-AD Model



The Basic AS-AD Model



Outline

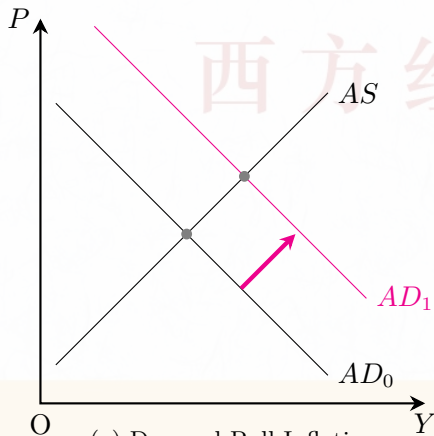
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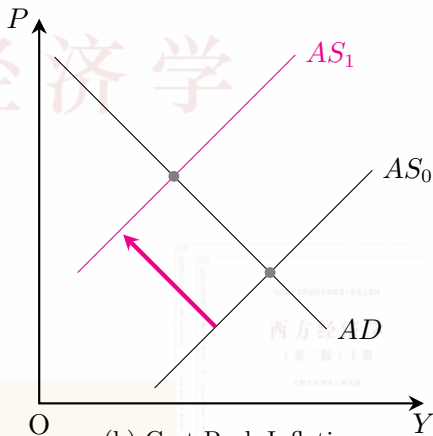


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Two Causes of Inflation



(a) Demand-Pull Inflation



(b) Cost-Push Inflation



Okun's Law in the United States

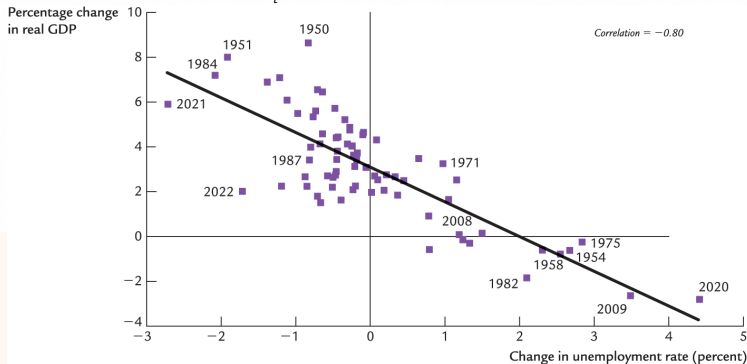
Okun's Law:

$$\ln(Y_t) - \ln(\bar{Y}_t) = -2(u_t - u_n).$$

The growth rate form:

$$g_{Y_t} - g_{\bar{Y}_t} = -2(u_t - u_{t-1}).$$

In the following figure, $g_{\bar{Y}_t} \approx 3$.



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中国石油大学
北京

The Phillips Curve

By means of $X_t - \bar{X} \approx \bar{X}(\ln X_t - \ln \bar{X})$,

$$\begin{aligned} Y_t &= \bar{Y} + \alpha \cdot (P_t - \mathbb{E}_{t-1} P_t) \\ \Rightarrow \ln P_t &= \mathbb{E}_{t-1} \ln P_t + \frac{1}{\tilde{\alpha}} (\ln Y_t - \ln \bar{Y}) \\ \Rightarrow \pi_t &= \mathbb{E}_{t-1} \pi_t + \frac{1}{\tilde{\alpha}} (\ln Y_t - \ln \bar{Y}). \end{aligned}$$

By Okun's law, $\ln Y_t - \ln \bar{Y} = -\tilde{\alpha}\beta(u_t - u_n)$. Substituting for $\ln Y_t - \ln \bar{Y}$ gives

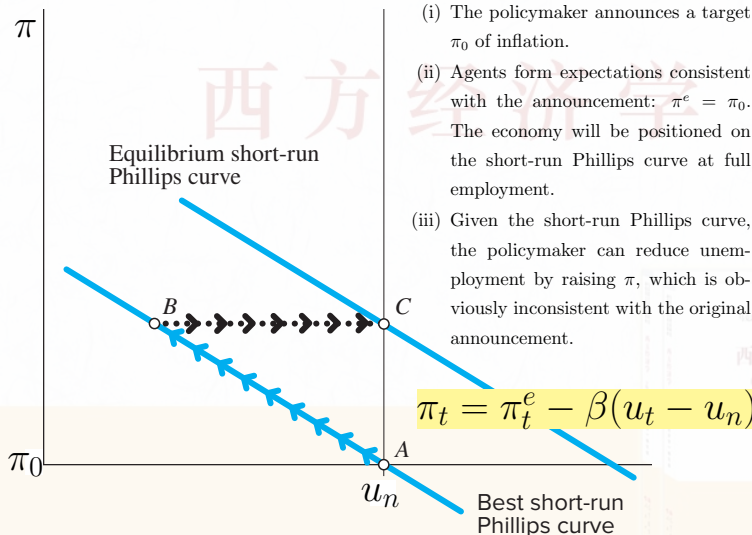
$$\pi_t = \pi_t^e - \beta(u_t - u_n),$$

which is called the **expectations-augmented Phillips curve**. If the supply shock v_t is considered, the Phillips curve becomes

$$\pi_t = \pi_t^e - \beta(u_t - u_n) + v_t,$$

The Phillips Curves

Dynamic/Time Inconsistency and the Long-Run Phillips Curve



See Dornbusch (2018)



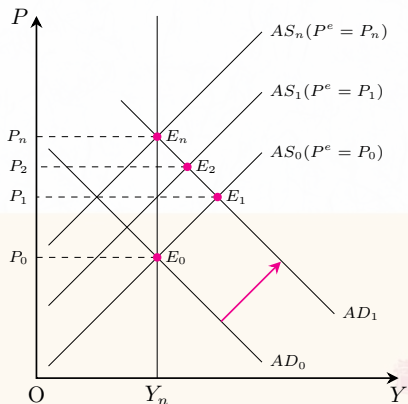
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Rational Expectations

Adaptive expectations are given by $P_t^e = P_{t-1}^e + \lambda(P_{t-1} - P_{t-1}^e)$.

Rational expectations assume that

- (1) People inside a model know the model;
- (2) People optimally use all the available information to forecast the future such that “outcomes do not differ systematically (i.e., regularly or predictably) from what people expected them to be.”



Proposition 1 (Policy Ineffectiveness)

Under the assumption of rational expectations, policies expected by people have no effect on output.

Proposition 2 (Lucas Critique)

Under the assumption of rational expectations, forecasts based only on historical information are invalid.

Disinflation with Pain

Traditional Approach to Disinflation

The Phillips curve is

$$\pi_t = \pi_t^e - \beta(u_t - u_n).$$

Suppose that an economy is initially in full employment, but with high inflation π^{High} . Suppose the government plans to lower the inflation from π^{High} to π^{Target} during T years. Given π_t^e , π_t falls at the cost of high u_t . Pain caused by disinflation can be measured by the sacrifice ratio.² In terms of output lost, the sacrifice ratio is defined as the number of percentage points of one year's real potential GDP that must be forgone to reduce inflation by 1 percentage point.

$$\text{Sacrifice Ratio } (SR_Y) = \frac{\text{Output Lost}}{\text{Decrease in inflation}} = \frac{\sum_{t=1}^T (\ln \bar{Y} - \ln Y_t)}{\pi^{High} - \pi^{Target}}.$$

In terms of unemployment tolerated, it is defined as the number of percentage points of one year's cyclical unemployment that must be tolerated to reduce inflation by 1 percentage point.

$$\text{Sacrifice Ratio } (SR_u) = \frac{\text{Cyclical Unemployment Tolerated}}{\text{Decrease in inflation}} = \frac{\sum_{t=1}^T (u_t - u_n)}{\pi^{High} - \pi^{Target}}.$$

²Another measure is the misery index, defined as $u + \pi$ (Dornbusch, 2018, ch6).

Disinflation without Pain

Rational Expectations Approach to Disinflation

The central bank makes a credible announcement that the money supply will decrease immediately and the inflation target is $\pi_t = \pi^{Target}$. The public will form $\pi_t^e = \pi^{Target}$. According to the Phillips curve,

$$\pi_t = \pi_t^e - \beta(u_t - u_n).$$

As a result, $\pi_t = \pi^{Target}$ while $u_t = u_n$. The essential ingredient of successful disinflation is the credibility of monetary policy.

Case 1 Zimbabwe's inflation hit about 100% a day in 2008. The hyperinflation stopped by April 2009 because it is legal for everyone to use U.S. dollars. (See ch22 of Dornbusch [2018, 13th ed.])

Case 2 Forward guidance (See [wiki](#), [FED](#))



Taylor Rule

In order to establish the credibility of monetary policy, the central bank is advised to follow a set of prespecified and publicly announced rules. Such an example is the *Taylor rule*,³ given by

$$i_t = \pi_t + r_n + \theta_\pi(\pi_t - \pi^{Target}) + \theta_Y \frac{Y_t - \bar{Y}}{\bar{Y}},$$

where $\theta_\pi > 0$ and $\theta_Y > 0$ are coefficients; i_t is the nominal interest rate controlled by the central bank; π_t is the rate of inflation; π^{Target} is the target rate of inflation; r_n is the natural rate of real interest; Y_t is the level of output; and $\bar{Y}$ is the natural level of output.

Proposition 3 (The Taylor Principle)

The central bank should respond to an increase in inflation with an even greater increase in the nominal interest rate. That is, $\frac{\partial i_t}{\partial \pi_t} > 1$.

Go to [FRED](#) to see the difference between effective federal funds rate and the rate implied by the Taylor rule.

³ [John Taylor](#) (1993, p.202) suggests a simple formula for monetary policy rule:

$$\text{The nominal Fed funds rate } i_t = \pi_t + 0.02 + 0.5 \times (\pi_t - 0.02) + 0.5 \times (Y_t - \bar{Y}) / \bar{Y}.$$

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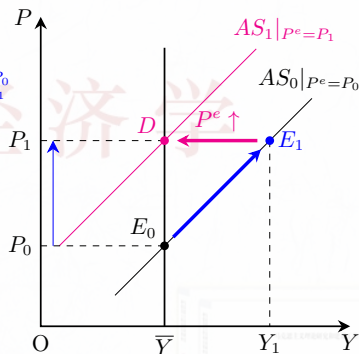
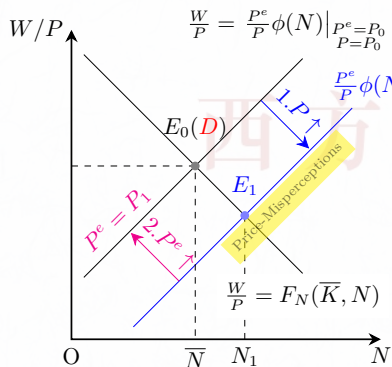
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Appendix: The Workers' Price-Misperceptions Model (Friedman, 1968)



In the short run:

$$P_0 \nearrow P_1 \Rightarrow \bar{N} \nearrow N_1, \bar{Y} \nearrow Y_1.$$

In the long run:

$$P^e \nearrow \Rightarrow N_1 \searrow \bar{N}, Y_1 \searrow \bar{Y}.$$

$$\bar{Y} = F(\bar{K}, \bar{N}), \quad Y_1 = F(\bar{K}, N_1).$$

Generally, the output is affected only by unperceived inflation.

Appendix: The Sticky-Price Model (Rotemberg, 1982)

When there are changes in demand due to changes in income, a fraction $s \in [0, 1]$ of firms charge sticky prices (P_{Sticky}) set in advance while a fraction $1 - s$ of firms charge flexible prices ($P_{Flexible}$). The general price level is $P = sP_{Sticky} + (1 - s)P_{Flexible}$. The price decision rules are

$$P_{Flexible} = P + a(Y - \bar{Y}), \quad a > 0;$$

$$P_{Sticky} = P^e + a(Y^e - \bar{Y}).$$

We assume that firms charging P_{Sticky} expect that the aggregate income, Y , will be equal to the natural level of output, $\bar{Y}$. Then the AS function can be written as

$$Y = \bar{Y} + \alpha(P - P^e)$$

where $\alpha = \frac{s}{(1-s)a} \geq 0$. AS is vertical if $s = 0$; horizontal if $s = 1$.



Appendix: The Imperfect-Information Model (Lucas, 1973)

There are n identical suppliers in an economy. Each supplier i can only observe one's own **flexible** price P_i . Due to imperfect information, each supplier cannot observe the general price level which is defined as $P = (\sum_{i=1}^n P_i)/n$ for simplicity. P^e is the expected general price level formed by supplier i based on all historical information. When supplier i observes his own price P_i , the expected general price level is revised to be $P^e|_{P_i}$.

The update rule: $P^e|_{P_i} = P^e + b(P_i - P^e), \quad b \in [0, 1];$

The production decision rule: $Y_i = \bar{Y}_i + a(P_i - P^e|_{P_i}), \quad a > 0,$

where $\bar{Y}_i$ is the output corresponding to $b = 1$ under perfect information. Let $Y = \sum_{i=1}^n Y_i$ and $\bar{Y} = \sum_{i=1}^n \bar{Y}_i$. We obtain

$$Y = \bar{Y} + \alpha(P - P^e)$$

where $\alpha = a(1 - b)n \geq 0$. This version of AS curve is called the *Lucas supply curve*.



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- (1) 掌握 AS 理论
- (2) 掌握 AS-AD 模型，并分析政策效应。
- (3) 什么是需求拉动型通胀？什么是成本推动型通胀？
- (4) 什么是 Okun's law?
- (5) 什么是 expectations-augmented Phillips curve?
- (6) 理解适应性预期和理性预期的概念。
- (7) 什么是 Lucas critique? 什么是 policy ineffectiveness?
- (8) 消除通胀有痛苦吗？有没有无痛消胀？什么是牺牲率？什么是前瞻性指引？
- (9) 什么是 Taylor rule?



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1 (E2, p.131)

根据马工程教材观点，应当如何评析西方经济学的 AS-AD 模型？

2 (E2, p.177)

根据马工程教材观点，应当如何评析西方经济学的菲利普斯曲线理论？



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